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J. DeSpirito, H.L. Edge, P. Weinacht, and J. Sahu
U.S. Army Research Laboratory
Aberdeen Proving Ground, MD

S.P.G. Dinavahi
Mississippi State University
Starkville, MS

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CFD ANALYSIS OF GRID FINS FOR MANEUVERING MISSILES

James DeSpirito,* Harris L. Edge,* Paul Weinacht,[†] and Jubaraj Sahu[‡]
U.S. Army Research Laboratory
Aberdeen Proving Ground, Maryland 21005

Surya P. G. Dinavahi[‡]
Mississippi State University
Starkville, Mississippi 39762

ABSTRACT

This paper presents the results of a study to demonstrate an approach for using viscous computational fluid dynamic simulations to calculate the flow field and aerodynamic coefficients for a missile with grid fins. A grid fin is an unconventional lifting and control surface that consists of an outer frame supporting an inner grid of intersecting planar surfaces of small chord. The calculations were made at a Mach number of 2.5 and several angles of attack for a missile without fins, with planar fins, and with grid fins. The results were validated by comparing the computed aerodynamic coefficients for the missile and individual grid fins against wind tunnel measurement data. Very good agreement with the measured data was observed for all configurations investigated. For the grid fin case, the aerodynamic coefficients were within 2.8–6.5% of the wind tunnel data. The normal force coefficient on the individual grid fins was within 11% of the test data. The simulations were also successful in calculating the flow structure around the fin in the separated-flow region at the higher angles of attack. This was evident in the successful calculation of the nonlinear behavior for that fin, which showed negative normal force at the higher angles of attack. The effective angle of attack is negative on part or the entire top grid fin for the higher angles of attack.

INTRODUCTION

The U.S. Army Research Laboratory (ARL) is using computational fluid dynamics (CFD) to investigate the aerodynamic characteristics and flow field structure of grid fins, also known as lattice controls. A grid fin is an unconventional lifting and control surface that consists of an outer frame supporting an inner grid of intersecting planar surfaces of small chord. Grid fins have been employed on some Soviet missiles but have

not been included on any missile systems in the West.¹ Interest in grid fins is primarily geared toward their potential use on highly maneuverable munitions due to their advantages over conventional planar controls at high angles of attack and high Mach number.

The aerodynamics of grid fins has been investigated since 1985 by U.S. Army Missile Research and Development Center (MRDEC), Huntsville, Alabama.^{2–4} These investigations indicated that grid fins have some advantages over conventional, planar fins. One advantage is the ability to maintain lift at higher angles of attack since grid fins do not have the same stall characteristics of planar fins. Another is the very small hinge moment, which can reduce the size of control actuator systems. Curvature of the grid fins had little effect on their performance so that folding the fins down onto the missile body is a storage design advantage. The main disadvantage was indicated to be higher drag than that of planar fins, although some techniques for minimizing drag by altering the grid fin frame cross-section shape were demonstrated.³ These studies also showed that grid fins experience a loss in control effectiveness in the transonic regime due to choking of the flow in the individual cells.

The Defence Evaluation and Research Agency (DERA), United Kingdom, has performed wind tunnel tests on grid fins and compared their aerodynamic characteristics to conventional planar fins.^{1,5} These studies confirmed some of the previous results and showed improved yaw stability due to the ability of the grid fins to generate side force. The results also showed that the vertical fins contribute about 30% of the static longitudinal stability when oriented in the cruciform (+) configuration.

Aeroballistic range flight tests have recently been conducted at the U.S. Air Force Research Laboratory Aeroballistic Research Facility (ARF), Eglin AFB, Florida.⁶ These were the first reported subscale free-flight tests of a missile configuration with grid fins. In addition to providing the raw data to derive the

*Aerospace Engineer, Senior Member

[†]Aerospace Engineer, Associate Fellow

[‡]Onsite CFD Lead, ARL MSRC PET Program, Senior Member

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aerodynamic coefficients for the grid fin models, shadowgraphs of the models in flight at Mach numbers between 0.39 and 1.65 showed an interesting shock structure in and around the fins.

There have also been theoretical and numerical methods used to estimate the lift characteristics of grid fins. Methods have been developed for the subsonic (vortex lattice theory), transonic (empirical methods), and supersonic (Evvard's theory) regimes.⁷⁻¹⁰ Empirical extensions were used to gain data for missiles at large angles of attack. These methods were concluded to provide adequate aerodynamic characteristics of grid fins for preliminary design purposes in the subsonic and supersonic regimes provided body upwash terms were included.

The first reported CFD calculations made on grid fins were sponsored by the Defence Research Establishment Valcartier (DREV), Canada.¹¹⁻¹³ These studies only included inviscid (Euler) simulations due mainly to a lack of computing resources. The authors concluded that the inviscid CFD calculations provided a reliable means of studying the flows past missiles with grid fins. Comparisons with data from Washington and Miller⁴ showed reasonable agreement for the fin normal force. It was noted that the Euler calculations for the grid fins compared better with the measured data than those for the planar fins. The authors noted that this could be attributed to the shorter surfaces of the grid fins in the axial direction, as compared to planar fins.¹² The results of the viscous calculations from the present study help to explain this effect.

The objective of the current investigation is to perform Navier-Stokes (N-S) simulations of a missile shape with grid fins and compare with experimental data. The results of this investigation are the first viscous CFD calculations to be made involving grid fins. Viscous solutions of grid fin missile aerodynamics allow a determination of the percentage of forces due to viscous and pressure effects. Comparison with inviscid solutions also determines the trade-off made if relying on inviscid calculations alone. This paper presents the results of N-S simulations of a generic missile shape with one grid fin configuration. Validation of the results is made by comparing against wind tunnel data provided by DERA.⁵ Simulations of the missile body alone and with planar fins were also performed as part of the validation process.

NUMERICAL APPROACH

The investigation involved using CFD to determine the flow field and aerodynamic coefficients on a 13-caliber, four-finned, generic missile shape (Figure 1). The missile has a 3-caliber tangent ogive, and the fin pitch axis is located 1.5 diameters before the aft end of the missile. The analysis proceeded in three steps: the first involved the missile without fins, case B1A (Figure 1, top); the second was with the missile with planar fins, case B1AC2R (Figure 1, middle); and the third was the missile with a set of grid fins, case B1AL2R (Figure 1, bottom). The configurations were labeled according to the designations used at DERA.⁵ The span and chord of the planar fin was 1.0 caliber. The span of the grid fin was 1.1 calibers and the chord was 0.17 calibers. All analyses were performed at a Mach number of 2.5 and at a minimum of three angles of attack (α): 0°, 10°, and 20°. The simulations were made of the missile in the cruciform (+) configuration, and symmetry (x-z plane) was used so that only a half-plane was modeled.

Steady-state calculations were made to compute the flow field for the three cases using the commercial CFD code, FLUENT Version 5.1.¹⁴ The implicit, compressible (coupled), unstructured-grid solver was used. The three-dimensional, time-dependent Reynolds-averaged Navier-Stokes (RANS) equations are solved using the finite volume method:

$$\frac{\partial}{\partial t} \int_V \mathbf{W} dV + \oint [\mathbf{F} - \mathbf{G}] \cdot d\mathbf{A} = \int_V \mathbf{H} dV, \quad (1)$$

where $\mathbf{W}$ is the vector of conservative variables, $\mathbf{F}$ and $\mathbf{G}$ are the inviscid and viscous flux vectors, respectively, $\mathbf{H}$ is the vector of source terms, V is the cell volume, and A is the surface area of the cell face. The Spalart-Allmaras,¹⁵ one-equation turbulence model was used for these calculations. In FLUENT, the original version of the Spalart-Allmaras model is modified to allow the use of wall functions when the mesh resolution is not sufficiently fine to resolve the viscous-affected, near-wall region of the boundary layer.¹⁶ This capability was used in generating the mesh so that the computational requirements were reduced as much as possible. Second-order discretization was used for the flow variables and the turbulent viscosity equation.

The geometry and unstructured meshes for the three configurations were generated using the preprocessor, GAMBIT, supplied in the FLUENT software suite. In generating the meshes, boundary layer mesh spacing

was used near the missile body and fin surfaces. Advantage was taken of the wall function option of the solver in FLUENT, and the first point off the surface (cell center) was about 0.002 calibers. All mesh stretching was kept below 1.2. Hexagonal cells were used except for a small region ahead of and partly over the first 0.1 calibers of the nose of the missile (1% of the total length). The latter region was made up of tetrahedrons and pyramid transition elements. Figure 2 shows the mesh on the symmetry plane for the B1A case, without fins. The triangular surface mesh can be observed at the upstream end. The tetrahedral mesh was made to cover a small part of the missile nose only to allow a transition between the two types of meshes near the nose. Therefore, a true boundary layer-type mesh was not covering the first 1% of the missile body, but this had no observable effect on the results. The B1A mesh in Figure 2 was meshed as six separate volumes—one in the freestream region, two over the first 10 calibers of the missile, and two over the last 3 calibers. This methodology was used so that the mesh in the first two regions could be used regardless of the fin type. The meshes between volumes were conformal, or exactly matching at the boundary surface. The total number of cells in this case was about 670,400.

The mesh on the symmetry plane for the planar fin case is shown in Figure 3. The mesh was modified slightly so that only one volume covered the first 10 calibers of the missile. The tail region, shown in Figure 4, consisted of seven volumes—ahead of, behind, and between the fins and off the fin tips. The total number of cells for this case was about 1.2 million, and again the mesh was totally conformal.

The geometry for the grid fin case, B1AL2R, is shown in Figure 5. The solid modeling capabilities of GAMBIT simplified the generation of the grid fin geometry. The mesh on the symmetry plane is shown in Figure 6. The mesh for the freestream region and the first 10 calibers of the missile body was similar to that for the planar fin case, Figure 3. Due to the complexity of the mesh in the region around the grid fins, and to minimize the total mesh size, a nonconformal mesh interface was used at the interface at 10 calibers. With this type of interface, which is used in FLUENT to handle sliding meshes, averaging of the flow variables is done to determine the value at the interface. The total number of cells in this mesh was about 3.2 million, with 2.5 million in the fin region (10–13 calibers). The mesh in the fin region is shown in Figure 7. The fins, interior cells, and the thin region circumferentially between the fins were meshed first. The surface mesh was then

projected axially forward and rearward with smoothing. A close-up of the surface mesh on the grid fin cells is shown in Figure 8. The number of cells across the front and rear of the grid fin web and frame surface was one or two. However, this dimension is the same order of magnitude as the first cell spacing off the surface (0.002 calibers).

The base flow was not simulated in these calculations, so the mesh ended at the end of the missile. An outflow boundary condition was used downstream, a pressure inflow (free-stream conditions) boundary condition was used upstream, and a far-field pressure (nonreflecting) boundary condition was used for the outer boundary. A nonslip wall boundary condition was used for all solid surfaces. The y^+ value was about 40–60 along the missile body, which is appropriate for wall functions. The maximum value was about 150 along the ogive and between 100 and 140 on the grid fin surfaces. The Reynolds number was 1.2 million.

The no fin and planar fin cases were also run with a structured-mesh, implicit, finite difference code, ZNSFLOW,¹⁷ that solved the thin-layer RANS equations. The aerodynamic coefficients calculated with this code, using the Baldwin-Lomax turbulence model, are presented in the next section along with those calculated with FLUENT. The grid fin simulations were originally to be made using ZNSFLOW and a Chimera overset grid.¹⁸ ZNSFLOW does not use wall functions in the turbulence models so that the first grid point of the solid surfaces was on the order of 5×10^{-6} calibers. The y^+ value was on the order of one in the ZNSFLOW calculations.

A series of inviscid calculations was also made on the grid fin configuration. These simulations used the same geometry as the viscous calculations except for the fin-body interface, where a small-length, circular stem was used for mesh generation purposes. A Cartesian mesh solver, TIGER,^{19,20} and automatic mesh generation program, CART3D,^{21,22} were used for the inviscid simulations. The mesh generation and solution time were much reduced over that required for the viscous calculations. The Cartesian grid had about 1.3 million grid points and the turnaround time was less than a day. The calculations for the flow solver were carried out on a Sun E10K machine on a single CPU. The flow solver took about 20 microseconds per cell per iteration and typically needed about 1,500 iterations for convergence. The aerodynamic coefficients

calculated with the inviscid simulations are compared to the viscous calculations in the following section.

The FLUENT simulations were performed in parallel using 4–6 processors on a Silicon Graphics (SGI) Onyx 2 with R12000 processors. The grid fin calculations took about 4.5 to 5 minutes per iteration using six processors. The aerodynamic coefficients converged in about 600 iterations. It took about 1,500 iterations for the turbulent viscosity to converge, with a reduction in the scaled residual to about 10^{-6} .

RESULTS AND DISCUSSION

Aerodynamic Coefficients

The viscous and pressure forces were integrated along the missile body and fin surfaces to calculate the aerodynamic coefficients. The normal force (C_z), axial force (C_x), and pitching moment (C_m) coefficients are presented in missile-based coordinates, as illustrated in Figure 9. The pitching moment is expressed about the nose of the missile. The reference area is the cross-sectional area of the missile base, and the reference length is the diameter of the missile. The calculated coefficients are compared to wind tunnel measurements performed at DERA.⁵ The DERA wind tunnel data did not include the forces on the base of the missile in the coefficient calculation. This value, which includes only the forebody and fin surfaces, is appropriate to compare to CFD calculations that don't include the base flow.

Body Alone Case (B1A). The aerodynamic coefficients calculated from the FLUENT and ZNSFLOW solutions are shown in Figures 10–12 as a function of α . Data from DERA wind tunnel measurements are also shown. The calculated pitching moment coefficient (Figure 10), normal force coefficient (Figure 11), and axial force coefficient (Figure 12), are shown at $\alpha = 0^\circ, 10^\circ, 14^\circ$, and 20° for the FLUENT calculations and at $0^\circ, 4^\circ, 10^\circ, 14^\circ$, and 20° for the ZNSFLOW calculations. The calculated results show very good agreement with the measured aerodynamic coefficients. The maximum difference between the calculated and measured pitching moment coefficient was 3.6% with the maximum occurring at $\alpha = 14^\circ$ for the FLUENT calculations, and 18% with a maximum at $\alpha = 10^\circ$ for the ZNSFLOW calculations. The maximum difference between the calculated and measured normal force coefficient was 1.9% with the maximum occurring at $\alpha = 10^\circ$ for the FLUENT calculations, and 10% with a maximum at $\alpha = 10^\circ$ for the ZNSFLOW calculations. The maximum difference between the calculated and measured axial force

coefficient was 4.0% with the maximum occurring at $\alpha = 20^\circ$ for the FLUENT calculations, and 12% with a maximum at $\alpha = 20^\circ$ for the ZNSFLOW calculations.

Planar Fin Case (B1AC2R). The aerodynamic coefficients calculated from the FLUENT and ZNSFLOW solutions are also shown in Figures 10–12 as a function of α . The calculated coefficients are shown at $\alpha = 0^\circ, 10^\circ$, and 20° for the FLUENT calculations and at $0^\circ, 4^\circ, 10^\circ, 14^\circ$, and 20° for the ZNSFLOW calculations. The calculated normal force and pitching moment coefficients again show very good agreement with the measured aerodynamic coefficients. The maximum difference between the calculated and measured pitching moment coefficient was 2.3% with the maximum occurring at $\alpha = 10^\circ$ for the FLUENT calculations, and 3.1% with a maximum at $\alpha = 14^\circ$ for the ZNSFLOW calculations. The maximum difference between the calculated and measured normal force coefficient was 0.8% with the maximum occurring at $\alpha = 10^\circ$ for the FLUENT calculations, and 1.5% with a maximum at $\alpha = 14^\circ$ for the ZNSFLOW calculations.

The calculated axial force coefficients were lower than the measured values but did predict the same increase with α that the experimental data showed. The FLUENT calculations were within 13–14% of the experimental data, and the ZNSFLOW calculations were within 17–22%. It is likely that the primary reason the FLUENT calculations were closer to the experimental data was the use of the one-equation turbulence model over the algebraic model used in ZNSFLOW. In addition, the thin-layer option used in the ZNSFLOW calculations eliminated the cross-stream turbulence effects in the fin region. It is interesting to note that the predicted values of axial force are lower than the measured values for this case. The opposite was found for the body alone case and the grid fin case, which is presented in the next section. Some further investigation is warranted, and it has been proposed to perform some CFD calculations of the planar fin model including part of the wind tunnel sting. In this way, the CFD predictions can be compared directly to the main balance force measured in the wind tunnel to eliminate the experimental base pressure modification as the cause for the discrepancy. The viscous component of the axial force was about 40% of the total, regardless of angle of attack, in the FLUENT and ZNSFLOW simulations.

Grid Fin Case (B1AL2R). The aerodynamic coefficients calculated from the FLUENT and the inviscid CART3D/TIGER solutions are shown in

Figures 13–15 as a function of α . The calculated coefficients are shown at $\alpha = 0^\circ, 10^\circ, 12^\circ$, and 20° for both CFD cases. The calculated normal force and pitching moment coefficients again show very good agreement with the measured aerodynamic coefficients. The maximum difference between the calculated and measured pitching moment coefficient was 6.2% with the maximum occurring at $\alpha = 12^\circ$ for the FLUENT calculations, and 18% with a maximum at $\alpha = 12^\circ$ for the CART3D/TIGER calculations. The maximum difference between the calculated and measured normal force coefficient was 6.5% with the maximum occurring at $\alpha = 12^\circ$ for the FLUENT calculations, and 13% with a maximum at $\alpha = 12^\circ$ for the CART3D/TIGER calculations.

The FLUENT axial force calculations were a little higher than the measured values: a 3–6% difference for $\alpha > 0$ and 11% at $\alpha = 0$. As would be expected, the inviscid CART3D/TIGER calculations underpredicted the axial force due to the absence of the skin friction. However, the difference (about 60%) was larger than expected. If the viscous component is removed from the axial force coefficient calculated in the FLUENT simulations, the value is between 0.34 and 0.38. The axial force coefficients calculated in the inviscid calculations are about one-half that (0.17–0.18). The viscous component of the axial force was 29% of the total, regardless of angle of attack, in the FLUENT simulations.

The inviscid calculations show that the total normal force and pitching moment data can be predicted to within 18% of the experimental data. If accurate axial force or drag information is not required, then the inviscid calculations may provide the information needed to check out multiple design approaches. Whether using CART3D/TIGER, or an inviscid solution with FLUENT, the reduction in computing time is substantial.

Forces on Fins. The normal force coefficients on the individual grid fins from the FLUENT calculations are shown in Figure 16, along with the measured wind tunnel data,⁵ as a function of α . The fins are numbered 1 to 4, with Fin 1 in the 3 o'clock position and Fin 4 in the 12 o'clock position if looking forward from the rear of the missile in the "+" configuration. In the simulations, Fin 1 and Fin 3 are the same due to symmetry. The normal force on the fins was predicted very well, with the largest difference about 11%. The normal force is greatest, as expected, on the horizontal fins. The windward fin (bottom, Fin 2) also provides a

substantial normal force. The leeward fin (top, Fin 4) provides a similar normal force as Fin 2 up to about 4° angle-of-attack but then goes nonlinear and negative at higher angles of attack. As described by Simpson,⁵ the nonlinear shape of the normal force vs. α curve for the leeward fin is due to its location in the separated flow region at higher angles of attack. As shown later in plots of the flowfield, the local angle of attack varies over the leeward grid fin. Some parts of the fin will be at an effective negative angle of attack, while other parts are at an effective positive angle of attack.

The axial force coefficients on individual grid fins were about 2–3 times that on the planar fins. The viscous component of the axial force on the grid fin was about 1.5 times that on the planar fins. Although it was speculated that the smaller chord of the grid fin might impart less viscous effects than a planar fin,¹² the summation of the viscous effects of all the lattice surfaces of the grid fin leads to higher viscous forces. The likely reason inviscid calculations of the aerodynamic coefficients on a missile with grid fins were more accurate than for a missile with planar fins¹² is that the viscous component of the total axial force on the fin is less for the grid fin case. In the present study, the viscous component of the axial force on the individual grid fin was 16% of the total versus 30% for the planar fin. With a larger component of the axial force in the grid fin case due to form and wave drag, an inviscid calculation may be expected to compare better with measured data than a planar fin case.

Grid Fin Flowfield

Contour plots of Mach number on the symmetry plane are shown for the 10° and 20° angle of attack cases in Figures 17 and 18, respectively. A strong oblique shock is seen emanating from the windward side of the nose, with a weaker shock coming off the leeward side. An expansion fan is seen coming off the ogive-body interface. A large separated-flow region is seen coming off the leeward side of the missile at $\alpha = 20^\circ$, enclosing the top grid fin. The separation zone is much smaller at $\alpha = 10^\circ$.

There is a complex, 3D-shock structure emanating off the grid fins. Figures 19 and 20 show pressure contours on the symmetry plane through the bottom and top fins, respectively, at $\alpha = 12^\circ$. The outline of the grid fin frame is shown in the figures, with the shocks emanating off the intersection of several grid fin cells (see Figures 7 and 8). At this Mach number, the shock and expansion waves do not reflect off the interior walls of the grid fin cells.² Instead, they first reflect off one

another inside the grid fin cell, setting up several more reflections in the wake of the fin. In Figures 21 and 22, contours of Mach number are shown on the bottom and top fins. For the bottom fin (Figure 21), the entire fin is at a positive relative angle of attack so that there is an expansion fan emanating from the lower part of each cell and a shock wave emanating off the upper part of each cell. For the top fin (Figure 22), part of the fin is at a negative relative angle of attack with respect to the incident flow due to the vortices in the separation zone. The top cell is at a positive angle of attack, with the shock wave emanating from the top of the cell. The cell second from the top is at a nearly 0° angle of attack, with shock waves emanating from the top and bottom of the cell. The third and fourth cells from the top are at an effective negative angle of attack, with the shock wave emanating from the bottom of the cell. At $\alpha = 20^\circ$, the entire top fin was at a negative relative angle of attack. This flow structure is responsible for the negative normal forces observed on the top fin in wind tunnel tests and these calculations (see Figure 16).

Contours of turbulent viscosity in the fin region are shown in Figures 23 and 24 for $\alpha = 10^\circ$ and 20° , respectively. Note that the scale in Figure 24 is larger than that in Figure 23. At $\alpha = 20^\circ$, there is more interaction of the separated flow coming off the missile body with the top grid fin.

SUMMARY AND CONCLUSIONS

Calculations of the viscous flow past a missile with grid fins were made using CFD simulations. The calculations were made at a Mach number of 2.5 and several angles of attack. The results were validated by comparing the computed aerodynamic coefficients for the missile and individual grid fins against detailed wind tunnel measurement data. The validation process also included calculating the flow field for the missile body alone and with conventional planar fins and comparing against wind tunnel data.

Very good agreement with the measured data was observed for all configurations investigated. For the grid fin case, the aerodynamic coefficients were within 6.5% of the wind tunnel data. The normal force coefficient on the individual grid fins was within 11% of the test data. The simulations were also successful in calculating the flow structure around the fin in the separated-flow region at the higher angles of attack. This was evident in the successful calculation of the nonlinear behavior for that fin, which showed negative normal force at the higher angles of attack. The physical nature of this effect was illustrated by

presenting contours of Mach number on the symmetry plane through the vertical grid fins. The effective angle of attack is negative on part or the entire top grid fin for the higher angles of attack.

Results for inviscid calculations of the grid fin case were also presented. The normal force and pitching moment coefficients were calculated to within 18% of the experimental data. If axial force or drag information is not required, then inviscid calculations may provide reasonable design data in less time than viscous calculations.

This study presented an approach for using viscous CFD simulations to calculate the flow field and aerodynamic coefficients for a missile with grid fins. Even when using an unstructured mesh and wall functions to reduce the mesh size and computational requirements, substantial computing resources were required. An alternative approach would be to use the Chimera overset grid technique to generate a structured mesh. However, the nature of the grid fin design makes the required resources large, regardless of the approach used.

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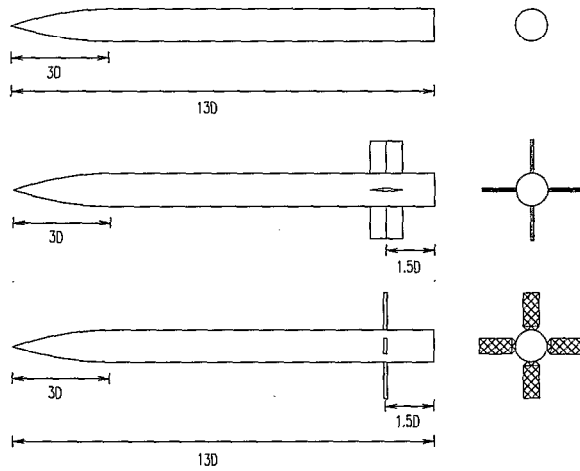


Figure 1. Basic missile shape with no fins (B1A, top), planar fins (B1AC2R, middle), and grid fins (B1AL2R, bottom).

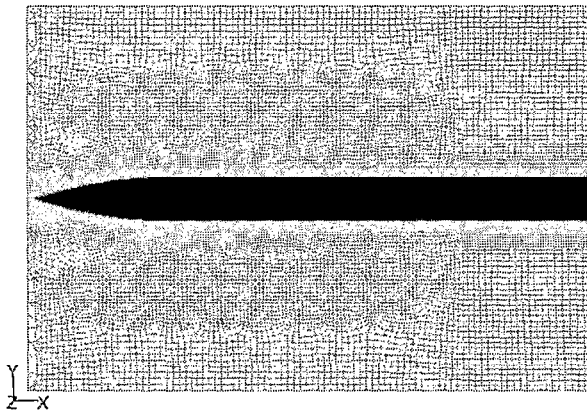


Figure 2. Unstructured mesh for basic missile shape without fins.

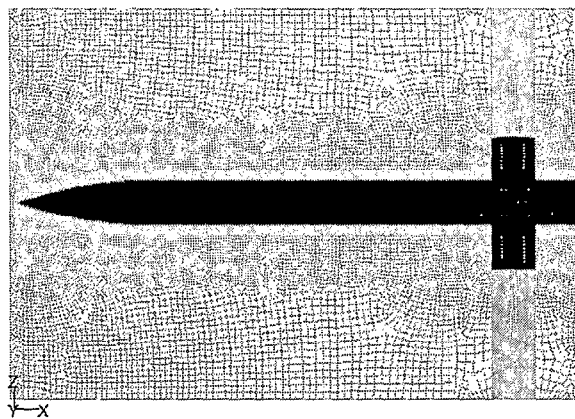


Figure 3. Unstructured mesh for basic missile shape with planar fins.

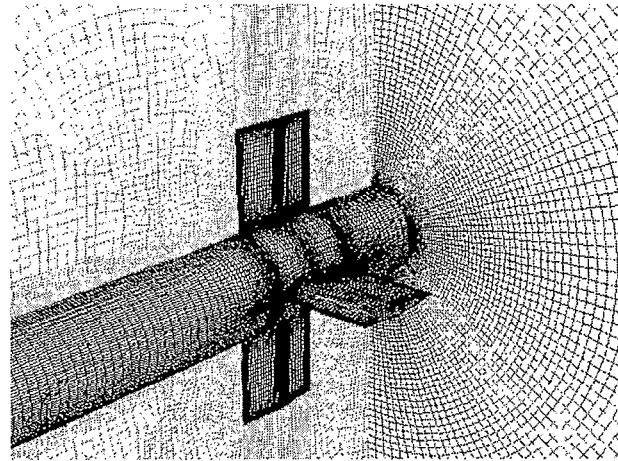


Figure 4. Unstructured mesh in tail region of planar fin case.

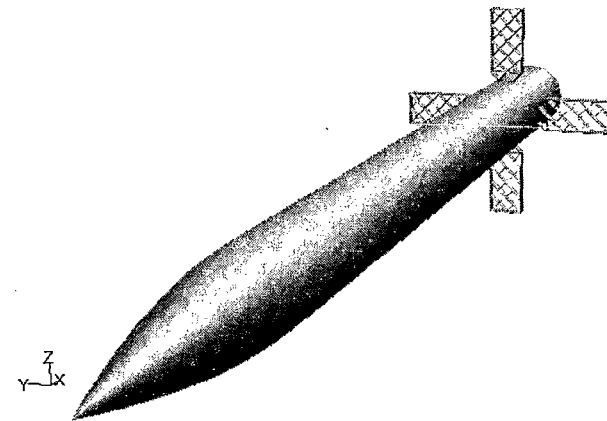


Figure 5. Geometry for grid fin case (B1AL2R).

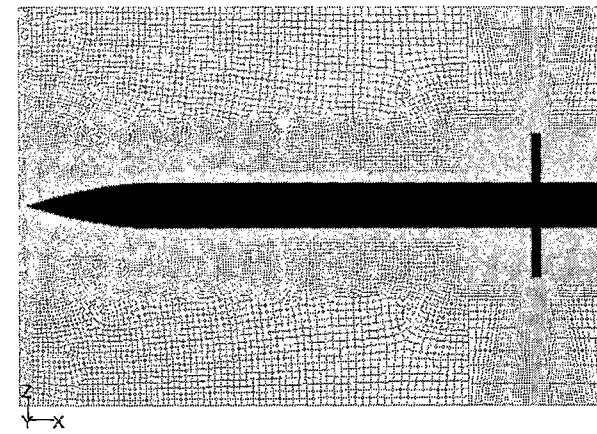


Figure 6. Unstructured mesh for basic missile shape with grid fins.

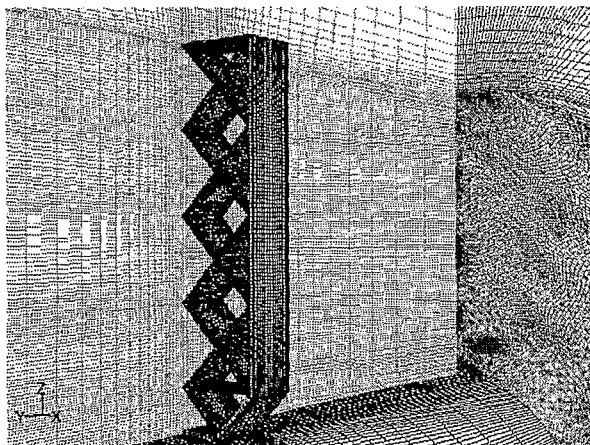


Figure 7. Unstructured mesh in tail region of grid fin case.

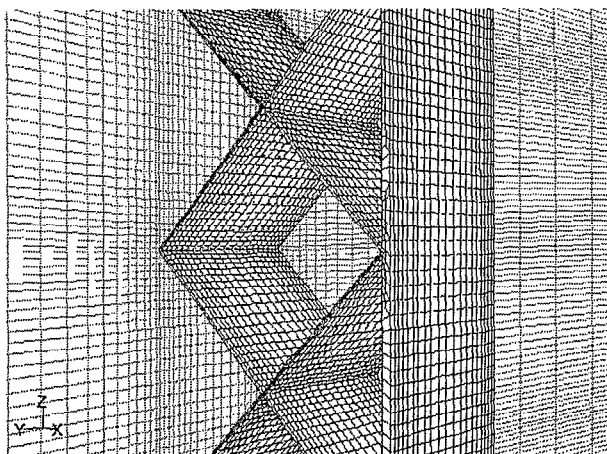


Figure 8. Surface mesh on grid fin cells.

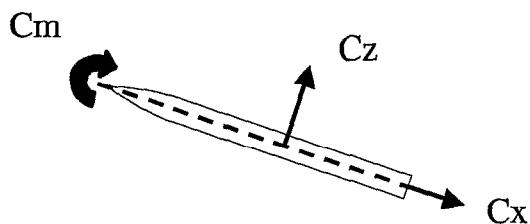


Figure 9. Axes and sign convention for force and moment coefficients.

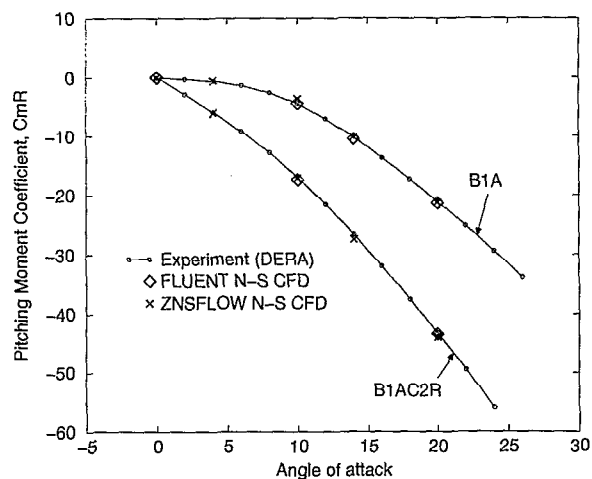


Figure 10. Pitching moment coefficient for B1A and B1AC2R cases.

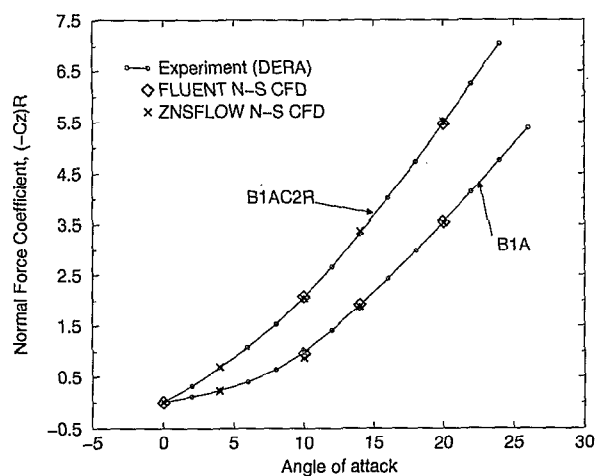


Figure 11. Normal force coefficient for B1A and B1AC2R cases.

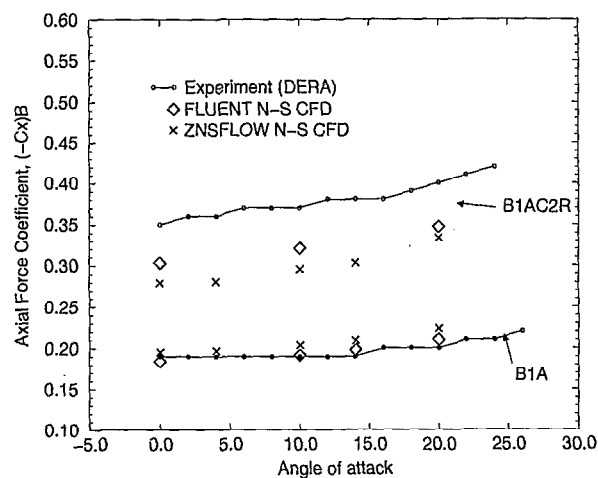


Figure 12. Axial force coefficient for B1A and B1AC2R cases.

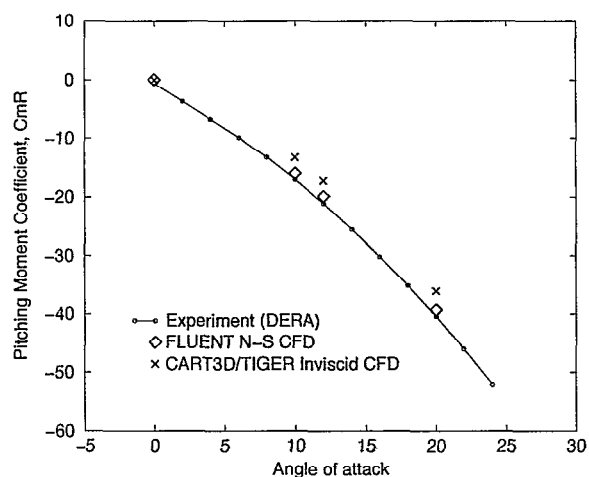


Figure 13. Pitching moment coefficient for B1AL2R case.

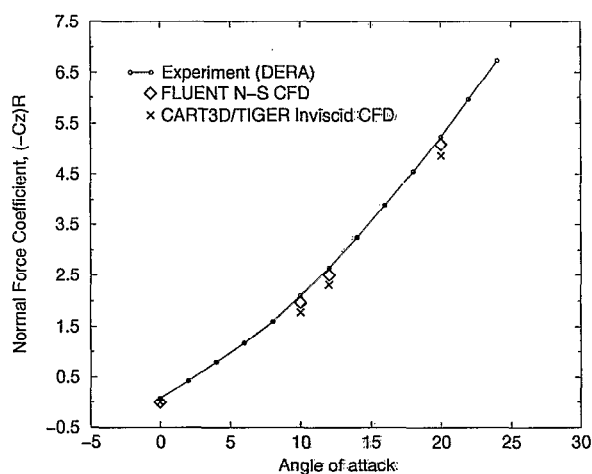


Figure 14. Normal force coefficient for B1AL2R case.

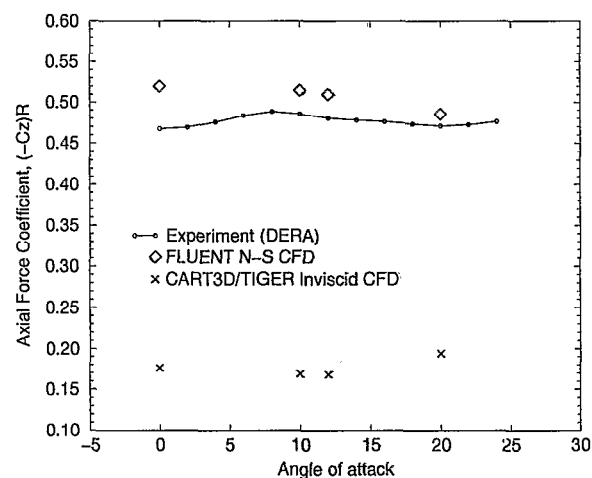


Figure 15. Axial force coefficient for B1AL2R case.

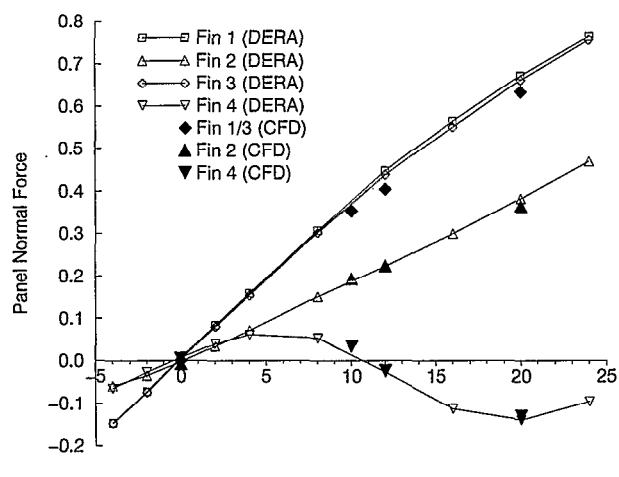


Figure 16. Normal force coefficient on individual grid fins.

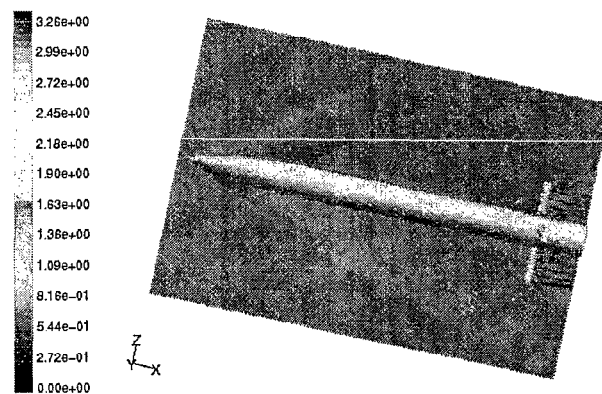


Figure 17. Mach number on symmetry plane for grid fin case, $\alpha = 10^\circ$.

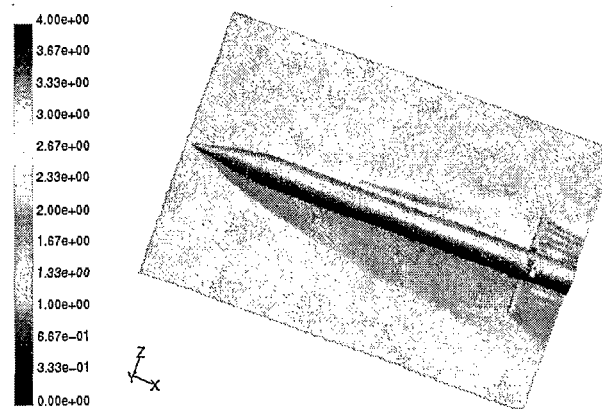


Figure 18. Mach number on symmetry plane for grid fin case, $\alpha = 20^\circ$.

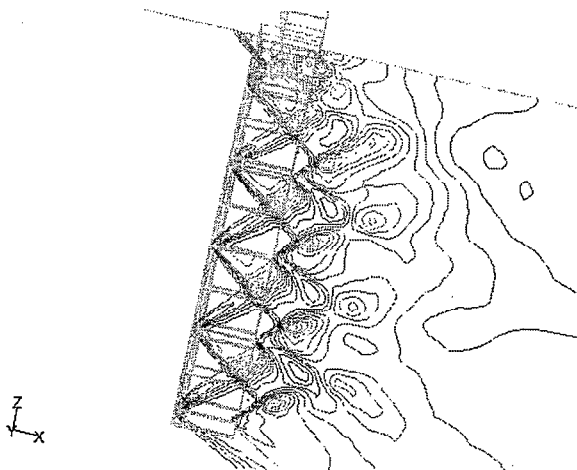


Figure 19. Pressure coefficient contours on symmetry plane through bottom fin (Fin 2).

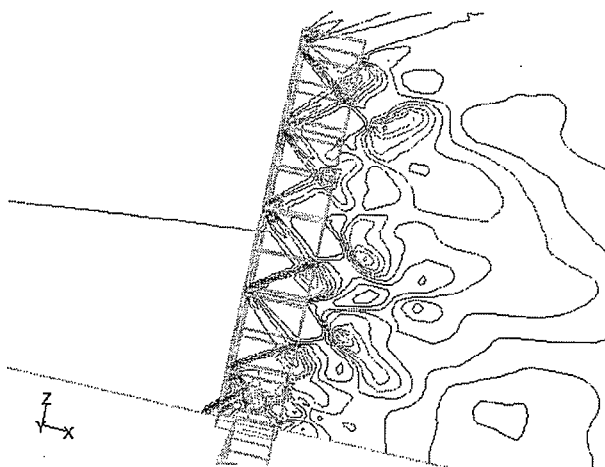


Figure 20. Pressure coefficient contours on symmetry plane through top fin (Fin 4).

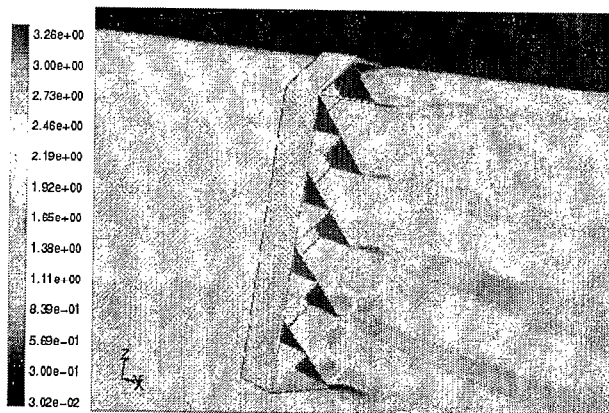


Figure 21. Mach number contours on symmetry plane through bottom fin (Fin 2).

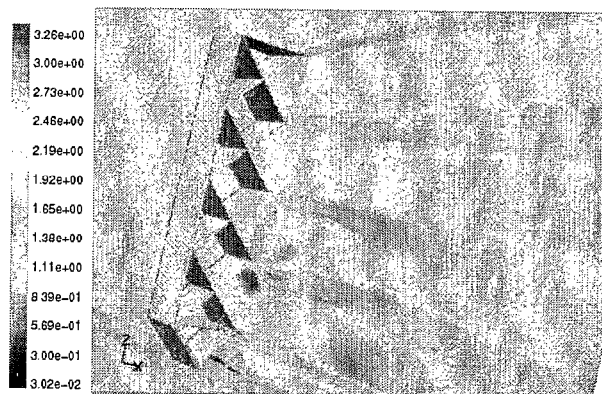


Figure 22. Mach number contours on symmetry plane through top fin (Fin 4).

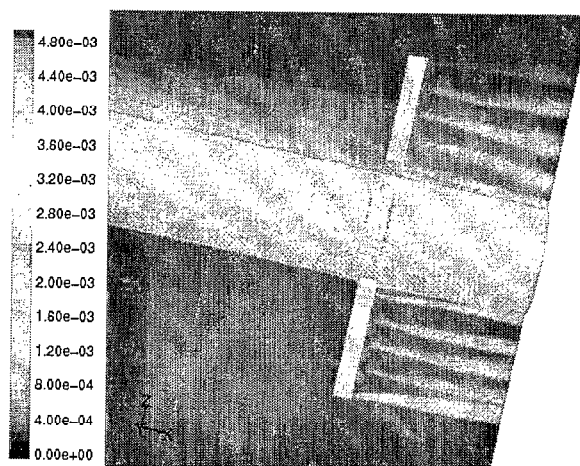


Figure 23. Turbulent viscosity contours on symmetry plane of fin region, $\alpha = 10^\circ$.

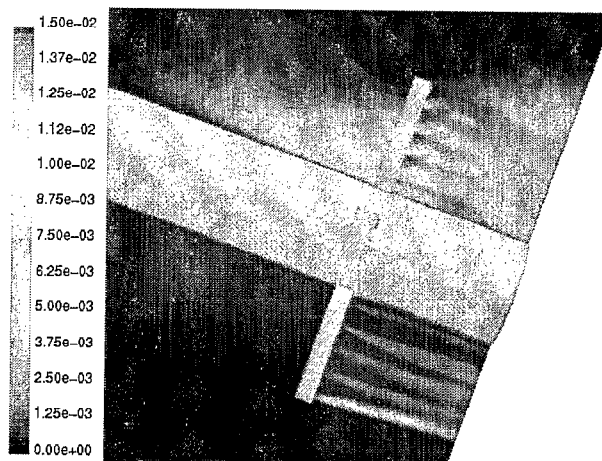


Figure 24. Turbulent viscosity contours on symmetry plane of fin region, $\alpha = 20^\circ$.